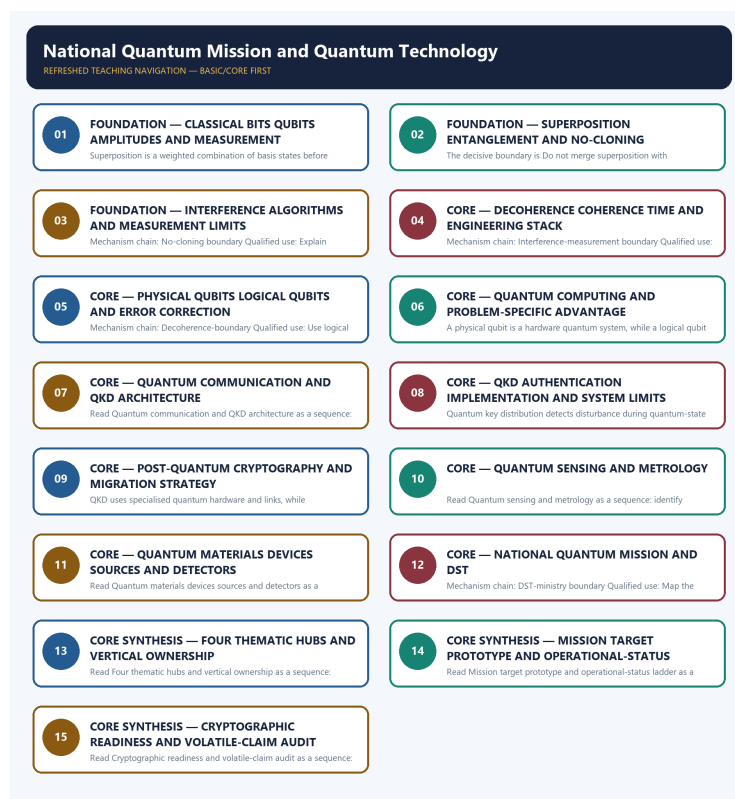


SOLVED PRACTICE WORKBOOK

National Quantum Mission and Quantum Technology - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Bit-qubit boundary?

A. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. B. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. C. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. D. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling.

Answer: A. Explanation: A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Bit-qubit boundary?

A. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. B. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. C. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. D. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Answer: B. Explanation: A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Bit-qubit boundary without changing its institution, unit or status?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. C. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. D. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Answer: C. Explanation: A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Bit-qubit boundary?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. C. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. D. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Answer: D. Explanation: A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Superposition-boundary?

A. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. B. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. C. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. D. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states.

Answer: A. Explanation: Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Superposition-boundary?

A. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. B. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. C. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. D. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Answer: B. Explanation: Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Superposition-boundary without changing its institution, unit or status?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. C. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. D. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count.

Answer: C. Explanation: Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Superposition-boundary?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. C. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. D. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement.

Answer: D. Explanation: Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Entanglement-boundary?

A. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. B. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. C. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. D. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Answer: A. Explanation: Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Entanglement-boundary?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. C. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. D. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Answer: B. Explanation: Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Entanglement-boundary without changing its institution, unit or status?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. C. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. D. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count.

Answer: C. Explanation: Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Entanglement-boundary?

A. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. B. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. C. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. D. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling.

Answer: D. Explanation: Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies No-cloning boundary?

A. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. B. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. C. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. D. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Answer: A. Explanation: The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of No-cloning boundary?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. C. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. D. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing.

Answer: B. Explanation: The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses No-cloning boundary without changing its institution, unit or status?

A. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. B. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. C. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. D. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing.

Answer: C. Explanation: The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about No-cloning boundary?

A. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. B. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. C. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. D. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states.

Answer: D. Explanation: The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Interference-measurement boundary?

A. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. B. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. C. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. D. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count.

Answer: A. Explanation: Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Interference-measurement boundary?

A. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. B. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. C. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. D. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing.

Answer: B. Explanation: Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Interference-measurement boundary without changing its institution, unit or status?

A. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. B. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. C. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. D. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing.

Answer: C. Explanation: Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Interference-measurement boundary?

A. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. B. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. C. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. D. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Answer: D. Explanation: Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Decoherence-boundary?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. C. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. D. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count.

Answer: A. Explanation: Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Decoherence-boundary?

A. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. B. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. C. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. D. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself.

Answer: B. Explanation: Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Decoherence-boundary without changing its institution, unit or status?

A. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. B. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. C. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. D. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself.

Answer: C. Explanation: Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Decoherence-boundary?

A. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. B. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. C. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. D. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints.

Answer: D. Explanation: Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Physical-logical boundary?

A. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. B. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. C. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. D. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer.

Answer: A. Explanation: A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Physical-logical boundary?

A. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. B. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. C. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. D. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself.

Answer: B. Explanation: A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Physical-logical boundary without changing its institution, unit or status?

A. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. B. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. C. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. D. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer.

Answer: C. Explanation: A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Physical-logical boundary?

A. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. B. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. C. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. D. A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count.

Answer: D. Explanation: A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Computing-boundary?

A. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. B. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. C. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. D. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself.

Answer: A. Explanation: Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Computing-boundary?

A. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. B. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. C. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. D. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration.

Answer: B. Explanation: Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Computing-boundary without changing its institution, unit or status?

A. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. B. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. C. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. D. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration.

Answer: C. Explanation: Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Computing-boundary?

A. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. B. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. C. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. D. Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing.

Answer: D. Explanation: Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Communication-boundary?

A. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. B. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. C. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. D. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration.

Answer: A. Explanation: Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Communication-boundary?

A. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. B. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. C. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. D. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration.

Answer: B. Explanation: Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Communication-boundary without changing its institution, unit or status?

A. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. B. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. C. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. D. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer.

Answer: C. Explanation: Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Communication-boundary?

A. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. B. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. C. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. D. Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer.

Answer: D. Explanation: Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies QKD-boundary?

A. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. B. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. C. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. D. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer.

Answer: A. Explanation: Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of QKD-boundary?

A. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. B. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. C. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. D. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing.

Answer: B. Explanation: Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses QKD-boundary without changing its institution, unit or status?

A. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. B. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. C. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. D. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing.

Answer: C. Explanation: Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about QKD-boundary?

A. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. B. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. C. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. D. Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself.

Answer: D. Explanation: Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies QKD-PQC boundary?

A. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. B. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. C. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. D. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer.

Answer: A. Explanation: QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of QKD-PQC boundary?

A. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. B. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. C. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. D. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry.

Answer: B. Explanation: QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses QKD-PQC boundary without changing its institution, unit or status?

A. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. B. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. C. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. D. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.

Answer: C. Explanation: QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about QKD-PQC boundary?

A. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. B. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. C. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. D. QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration.

Answer: D. Explanation: QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Sensing-metrology boundary?

A. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. B. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. C. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. D. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing.

Answer: A. Explanation: Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Sensing-metrology boundary?

A. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. B. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. C. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. D. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.

Answer: B. Explanation: Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Sensing-metrology boundary without changing its institution, unit or status?

A. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. B. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. C. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. D. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.

Answer: C. Explanation: Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Sensing-metrology boundary?

A. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. B. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. C. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. D. Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer.

Answer: D. Explanation: Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Materials-devices boundary?

A. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. B. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. C. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. D. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product.

Answer: A. Explanation: Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Materials-devices boundary?

A. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. B. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. C. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. D. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs.

Answer: B. Explanation: Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Materials-devices boundary without changing its institution, unit or status?

A. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. B. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. C. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. D. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.

Answer: C. Explanation: Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Materials-devices boundary?

A. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. B. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. C. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. D. Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing.

Answer: D. Explanation: Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies NQM-mission boundary?

A. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. B. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. C. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. D. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs.

Answer: A. Explanation: The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of NQM-mission boundary?

A. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. B. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. C. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. D. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved.

Answer: B. Explanation: The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses NQM-mission boundary without changing its institution, unit or status?

A. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. B. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. C. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. D. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists.

Answer: C. Explanation: The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about NQM-mission boundary?

A. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. B. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. C. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. D. The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product.

Answer: D. Explanation: The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies DST-ministry boundary?

A. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. B. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. C. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. D. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.

Answer: A. Explanation: The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of DST-ministry boundary?

A. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. B. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. C. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. D. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved.

Answer: B. Explanation: The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses DST-ministry boundary without changing its institution, unit or status?

A. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. B. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. C. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. D. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved.

Answer: C. Explanation: The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about DST-ministry boundary?

A. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. B. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. C. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. D. The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry.

Answer: D. Explanation: The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies Four-hub boundary?

A. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. B. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. C. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. D. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved.

Answer: A. Explanation: NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of Four-hub boundary?

A. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. B. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. C. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. D. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements.

Answer: B. Explanation: NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses Four-hub boundary without changing its institution, unit or status?

A. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. B. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. C. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. D. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements.

Answer: C. Explanation: NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an

operational system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about Four-hub boundary?

A. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. B. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. C. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. D. NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.

Answer: D. Explanation: NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Target-achievement boundary?

A. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. B. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. C. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. D. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements.

Answer: A. Explanation: A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Target-achievement boundary?

A. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. B. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. C. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. D. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Answer: B. Explanation: A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Target-achievement boundary without changing its institution, unit or status?

A. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. B. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. C. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. D. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Answer: C. Explanation: A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Target-achievement boundary?

A. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. B. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. C. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. D. A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs.

Answer: D. Explanation: A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Qubit-target boundary?

A. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. B. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. C. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. D. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists.

Answer: A. Explanation: The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Qubit-target boundary?

A. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. B. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. C. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. D. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Answer: B. Explanation: The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Qubit-target boundary without changing its institution, unit or status?

A. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. B. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. C. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. D. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling.

Answer: C. Explanation: The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Qubit-target boundary?

A. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. B. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. C. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. D. The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved.

Answer: D. Explanation: The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Cryptographic-readiness boundary?

A. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. B. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. C. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. D. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Answer: A. Explanation: Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Cryptographic-readiness boundary?

A. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. B. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. C. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. D. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement.

Answer: B. Explanation: Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Cryptographic-readiness boundary without changing its institution, unit or status?

A. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. B. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. C. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. D. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling.

Answer: C. Explanation: Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Cryptographic-readiness boundary?

A. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. B. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. C. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. D. Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists.

Answer: D. Explanation: Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile-claim boundary?

A. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. B. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. C. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. D. A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Answer: A. Explanation: Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile-claim boundary?

A. Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. B. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. C. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. D. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling.

Answer: B. Explanation: Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile-claim boundary without changing its institution, unit or status?

A. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. B. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. C. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. D. Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling.

Answer: C. Explanation: Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile-claim boundary?

A. Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. B. The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. C. Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. D. Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements.

Answer: D. Explanation: Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2022 qubit concept, the 2025 Majorana-company-claim taxonomy and a provisional 2026 NQM-target concept here. No objective key, achieved qubit count, communication distance or operational capability is invented.

PYQ DEMAND CARD 1 - 2022 Prelims GS-I

Demand: Identify the qubit concept in a quantum-computing context.

Status: Verified routed objective concept; the official key was unavailable locally, so no answer letter is asserted.

Model solution: Bit-qubit boundary: A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Superposition-boundary: Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement.

Entanglement-boundary: Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. **Interference-measurement boundary:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat "PYQ DEMAND CARD 1 - 2022 Prelims GS-I" as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Bit-qubit boundary: A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.

Superposition-boundary: Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement.

Entanglement-boundary: Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. **Interference-measurement boundary:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

- 1. Claim and named evidence:** Demand: Identify the qubit concept in a quantum-computing context. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
- 2. Claim and named evidence:** Status: Verified routed objective concept; the official key was unavailable locally, so no answer letter is asserted. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Bit-qubit boundary: A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. **Superposition-boundary:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Entanglement-boundary:** Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. **Interference-measurement boundary:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2022 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 2 - 2025 Prelims GS-I

Demand: Assess claims about the Majorana 1 chip, quantum computing and deep learning.

Status: Verified routed objective concept; the package preserves company-claim and technology-taxonomy boundaries without asserting an answer letter.

Model solution: Physical-logical boundary: A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Computing-boundary:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Target-achievement boundary:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Volatile-claim boundary:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 2 - 2025 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: Physical-logical boundary: A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Computing-boundary:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Target-achievement boundary:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Volatile-claim boundary:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess claims about the Majorana 1 chip, quantum computing and deep learning. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Status: Verified routed objective concept; the package preserves company-claim and technology-taxonomy boundaries without asserting an answer letter. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Physical-logical boundary: A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Computing-boundary:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Target-achievement boundary:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Volatile-claim boundary:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For “PYQ DEMAND CARD 2 - 2025 Prelims GS-I”, explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

PYQ DEMAND CARD 3 - 2026 Prelims GS-I

Demand: Assess National Quantum Mission targets and thematic hubs.

Status: Provisional routed concept; no option letter, achieved qubit count or operational system is inferred.

Model solution: NQM-mission boundary: The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **DST-ministry boundary:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Four-hub boundary:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.

Target-achievement boundary: A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs.

Qubit-target boundary: The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Volatile-claim boundary:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: Treat “PYQ DEMAND CARD 3 - 2026 Prelims GS-I” as a mechanism, category, institution, unit, date and status problem.

Detailed examiner-grade model answer:

Introduction and thesis: NQM-mission boundary: The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **DST-ministry boundary:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Four-hub boundary:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational

system. **Target-achievement boundary:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Qubit-target boundary:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Volatile-claim boundary:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Assess National Quantum Mission targets and thematic hubs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Status: Provisional routed concept; no option letter, achieved qubit count or operational system is inferred. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: NQM-mission boundary: The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **DST-ministry boundary:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Four-hub boundary:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. **Target-achievement boundary:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Qubit-target boundary:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Volatile-claim boundary:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: Identify the object and actor; trace the technical mechanism; fix the measurement and platform; test each statement against source date, status rung and closest exception.

Why this earns marks: It prevents organisation, platform, process, application, regulatory role, target and observed outcome from being conflated.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2026 Prelims GS-I", explain why the nearest distractor fails on mechanism, platform, unit, institution, legal status, maturity or date.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish a classical bit from a qubit and explain measurement. Answer in about 150 words.

Model thesis: Claim: Bit-qubit boundary. **Named evidence/example:** A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit,

institution, system category, legal character and development-to-deployment status boundary. **Claim:** Interference-measurement boundary. **Named evidence/example:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit.
- Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement.
- Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results.

Qualified conclusion: Claim: Bit-qubit boundary. **Named evidence/example:** A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Interference-measurement boundary. **Named evidence/example:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Distinguish a classical bit from a qubit and explain measurement. Answer in about 150 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Bit-qubit boundary. **Named evidence/example:** A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Interference-measurement boundary. **Named evidence/example:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Bit-qubit boundary. **Named evidence/example:** A classical bit is recorded as 0 or 1, while a qubit is a quantum state described by amplitudes over basis states and yields a classical outcome on measurement; it is not simply a faster bit. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: Interference-measurement boundary. **Named evidence/example:** Quantum algorithms manipulate amplitudes so interference can suppress unwanted paths and reinforce useful outcomes before measurement; superposition alone does not test every answer and reveal all results. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Distinguish a classical bit from a qubit and explain measurement. Answer in about 150 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate superposition, entanglement and no-cloning. Answer in about 150 words.

Model thesis: **Claim:** Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Entanglement-boundary. **Named evidence/example:** Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement.
- Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling.
- The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states.

Qualified conclusion: **Claim:** Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Entanglement-boundary. **Named evidence/example:** Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **answer** requires a direct position on "Differentiate superposition, entanglement and no-cloning. Answer in about 150 words.", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Entanglement-boundary. **Named evidence/example:** Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement

enables no faster-than-light signalling. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Superposition-boundary. **Named evidence/example:** Superposition is a weighted combination of basis states before measurement; saying a qubit is two ordinary classical values at once obscures amplitudes, probability and measurement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Entanglement-boundary. **Named evidence/example:** Entanglement is a joint non-factorisable state of two or more quantum systems; a single qubit can be in superposition but cannot be entangled with itself, and entanglement enables no faster-than-light signalling. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Differentiate superposition, entanglement and no-cloning. Answer in about 150 words.", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain why physical-qubit counts do not establish fault-tolerant quantum capability. Answer in about 250 words.

Model thesis: **Claim:** Decoherence-boundary. **Named evidence/example:** Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physical-logical boundary. **Named evidence/example:** A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints.
- A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count.
- The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved.

Qualified conclusion: **Claim:** Decoherence-boundary. **Named evidence/example:** Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physical-logical boundary. **Named evidence/example:** A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **explain** requires a direct position on "Explain why physical-qubit counts do not establish fault-tolerant quantum capability. Answer...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Decoherence-boundary. **Named evidence/example:** Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physical-logical boundary. **Named evidence/example:** A physical qubit is a hardware quantum system, while

a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Decoherence-boundary. **Named evidence/example:** Decoherence is loss of usable quantum coherence through environmental interaction, making isolation, cryogenics, vacuum, shielding, control and materials central engineering constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Physical-logical boundary. **Named evidence/example:** A physical qubit is a hardware quantum system, while a logical qubit is error-corrected information encoded across many physical qubits; a physical-qubit count is not a fault-tolerant capability count. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Explain why physical-qubit counts do not establish fault-tolerant quantum capability. Answer...", replace the weakest catalogue point with one exact mechanism, named

mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Compare quantum computing, communication, sensing and materials as separate technology families. Answer in about 250 words.

Model thesis: Claim: Computing-boundary. **Named evidence/example:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sensing-metrology boundary. **Named evidence/example:** Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Materials-devices boundary. **Named evidence/example:** Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing.
- Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer.
- Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer.
- Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing.

Qualified conclusion: Claim: Computing-boundary. **Named evidence/example:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sensing-metrology boundary. **Named evidence/example:** Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Materials-devices boundary. **Named evidence/example:** Quantum materials and devices provide sources,

detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **compare** requires a direct position on "Compare quantum computing, communication, sensing and materials as separate technology...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Computing-boundary. **Named evidence/example:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sensing-metrology boundary. **Named evidence/example:** Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Materials-devices boundary. **Named evidence/example:**

Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Computing-boundary. **Named evidence/example:** Quantum computing offers different algorithms for selected problem classes such as simulation, factoring, search or optimisation; it is not universally faster and does not replace ordinary classical computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Sensing-metrology boundary. **Named evidence/example:** Quantum sensing and metrology exploit quantum sensitivity for precise measurement such as clocks, magnetic or gravitational fields; a quantum sensor is not a smaller general-purpose quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Materials-devices boundary. **Named evidence/example:** Quantum materials and devices provide sources, detectors, control components and hardware substrates for computing, communication and sensing; the vertical is not a synonym for quantum computing. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Compare quantum computing, communication, sensing and materials as separate technology...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate the National Quantum Mission through its four hubs, targets and readiness boundaries. Answer in about 300 words.

Model thesis: Claim: NQM-mission boundary. **Named evidence/example:** The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST-ministry boundary. **Named evidence/example:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Four-hub boundary. **Named evidence/example:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Target-achievement boundary. **Named evidence/example:** A mission target states intended capability within a

period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product.
- The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry.
- NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system.
- A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs.
- The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved.
- Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements.

Qualified conclusion: **Claim:** NQM-mission boundary. **Named evidence/example:** The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST-ministry boundary. **Named evidence/example:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Four-hub boundary. **Named evidence/example:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Target-achievement boundary. **Named evidence/example:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution,

system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary.

Named evidence/example: Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **evaluate** requires a direct position on "Evaluate the National Quantum Mission through its four hubs, targets and readiness...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: NQM-mission boundary. **Named evidence/example:** The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST-ministry boundary. **Named evidence/example:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Four-hub boundary. **Named evidence/example:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Target-achievement boundary. **Named evidence/example:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

3. **Claim and named evidence:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: NQM-mission boundary. **Named evidence/example:** The National Quantum Mission is a DST-led capability mission across computing, communication, sensing and metrology, and materials and devices; it is not one machine, laboratory or company product. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DST-ministry boundary. **Named evidence/example:** The Department of Science and Technology is a department under the Ministry of Science and Technology and implements NQM; DST should not be called a separate ministry. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Four-hub boundary. **Named evidence/example:** NQM uses four thematic hubs hosted at IISc Bengaluru, IIT Madras with C-DOT, IIT Bombay and IIT Delhi for the four distinct verticals; a hub is an ecosystem institution, not proof of an operational system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Target-achievement boundary. **Named evidence/example:** A mission target states intended capability within a period, while approved funding, a hub, a research prototype, a laboratory demonstration and an operational system are separate evidence rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Qubit-target boundary. **Named evidence/example:** The NQM physical-qubit target is a hardware-scaling objective, not a claim that logical error-corrected qubits, useful quantum advantage or fault-tolerant computing have been achieved. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit,

institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Evaluate the National Quantum Mission through its four hubs, targets and readiness...”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Assess India's quantum-security strategy by distinguishing QKD, post-quantum cryptography and cryptographic readiness. Answer in about 300 words.

Model thesis: Claim: No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** QKD-boundary. **Named evidence/example:** Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** QKD-PQC boundary. **Named evidence/example:** QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cryptographic-readiness boundary. **Named evidence/example:** Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states.
- Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer.

- Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself.
- QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration.
- Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists.
- Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements.

Qualified conclusion: Claim: No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: QKD-boundary. **Named evidence/example:** Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** QKD-PQC boundary. **Named evidence/example:** QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cryptographic-readiness boundary. **Named evidence/example:** Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Demand decoding: The directive **assess** requires a direct position on "Assess India's quantum-security strategy by distinguishing QKD, post-quantum cryptography and...", each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution,

application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: QKD-boundary. **Named evidence/example:** Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** QKD-PQC boundary. **Named evidence/example:** QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cryptographic-readiness boundary. **Named evidence/example:** Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Analytical body:

1. **Claim and named evidence:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** Connect mechanism -> named system/institution -> application or

implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: No-cloning boundary. **Named evidence/example:** The no-cloning principle forbids perfect copying of an arbitrary unknown quantum state; it helps explain quantum-communication security but does not prevent all copying of classical information or known basis states. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Communication-boundary. **Named evidence/example:** Quantum communication uses quantum states and protocols for tasks such as key distribution, while quantum computing processes information; a secure link is not a quantum computer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim: QKD-boundary. **Named evidence/example:** Quantum key distribution detects disturbance during quantum-state exchange for key establishment, but still needs authenticated endpoints, implementation security and a wider encryption system; QKD does not encrypt all data by itself. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** QKD-PQC boundary. **Named evidence/example:** QKD uses specialised quantum hardware and links, while post-quantum cryptography uses classical mathematical algorithms designed to resist quantum attack and can be deployed through software and standards migration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cryptographic-readiness boundary. **Named evidence/example:** Quantum risk includes harvest-now-decrypt-later exposure, so cryptographic inventory and post-quantum migration are present governance tasks even before a cryptographically relevant quantum computer exists. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Volatile-claim boundary. **Named evidence/example:** Mission outlay, dates, target qubits, communication distance, startup count, prototype performance and operational status require dated official DST or mission sources and cannot be upgraded from announcements. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For "Assess India's quantum-security strategy by distinguishing QKD, post-quantum cryptography and...", replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.